

LED Matrix Animation Player

Step-by-Step Operator Guide & Troubleshooting Manual

1. Introduction & Quick Start

The **LED Matrix Animation Player** (animation_viewer.html) is a standalone browser web application designed to preview, verify, and analyze raw .bin animation files for WS2812B / NeoPixel LED matrices without requiring an ESP32 hardware connection.

Step	Action	Details
1	Open Web Application	Double-click animation_viewer.html in any web browser.
2	Load Animation File	Drag and drop your .bin file into the upload zone or click to select.
3	Configure Dimensions	Use ■ Run Auto-Detect or enter the Width x Height manually.
4	Play & Adjust	Click Play , scrub frames, and adjust playback speed.

2. Setting Matrix Dimensions: Auto-Detect vs. Manual

To display an animation accurately, the player needs to know the matrix dimensions (Width in columns x Height in rows). You can set these using either Auto-Detect or Manual Input:

A. Auto-Detect Dimensions (Recommended First Step)

- Click **■ Run Auto-Detect** in the blue sidebar panel.
- **How Auto-Detect works:** It inspects filename patterns (e.g. 25X20-ISLAMI_0_orig.bin $\rightarrow$ 25x20), parses binary header metadata bytes, and resolves LED factor ratios.
- **When it works:** Ideal for standard exported files with descriptive filenames or embedded metadata headers.
- **Note:** If the file is headerless or has a generic name (like data.bin), Auto-Detect may suggest a default 25x20 ratio. In this case, use Manual Input.
- Click **✓ Apply Detected Dimensions** to apply the recommendation.

B. Manual Matrix Input (Always Available)

- You can directly type the **Width (Columns)** and **Height (Rows)** at any time.
- Or choose from common hardware presets in the **Presets** dropdown (e.g., 20x25, 25x20, 32x32, 64x64, 72x72, 120x150).
- **Important:** Manual inputs are never removed and will always override auto-detection when modified.

3. Binary Decoding & Alignment Options

Different LED animation software tools export binary files using varying offsets, color orders, and wiring paths. The sidebar contains controls to align any format perfectly:

Setting	Options	When to Use
Header Offset	• 0 Bytes (Default) • 512 Bytes • Auto-Detect	Set to 0 Bytes for raw headerless files. Set to 512 Bytes for files exported from tools like Jinx!/Glediator that include 512B metadata headers.
Color Order	• RGB (Standard) • GRB (WS2812B) • BGR (Reversed)	Use RGB for standard display. Switch to GRB if red and green colors appear swapped during playback.
Wiring Layout	• Progressive • Serpentine (Zigzag)	Use Progressive if rows run left-to-right. Use Serpentine if physical wiring zigzags back and forth every row.
Start Corner	• Top-Left (Default) • Bottom-Left • Top-Right / Bottom-Right	Adjust if the animation appears upside-down or horizontally mirrored.
Rotation	• 0° / 90° / 180° / 270° •  Rotate 90° Button	Rotates the canvas view in real time without reloading the file.

4. Playback Controls & Speed Adjustment

- **Play / Pause:** Toggle animation playback.
- **Timeline Scrubber:** Drag to scrub to any frame in the file.
- **Step Controls:** Use Step -1 and Step +1 for precise frame-by-frame inspection.
- **Speed Controls:** Use the **Speed Dropdown** on the player bar for quick presets (0.25x Slow Mo, 0.5x, 1.0x Normal, 2.0x, 4.0x Ultra Fast) or the **FPS Slider** (1-60 FPS) in the sidebar.
- **Pixel Size & Glow:** Adjust pixel size slider and toggle NeoPixel glow effect for visual realism.

5. Troubleshooting Visual Distortion Issues

Symptom / Visual Issue	Root Cause	How to Fix
Scrambled, diagonal, or skewed lines	Header offset or matrix dimensions mismatch	Toggle Header Offset between 0 Bytes and 512 Bytes , or verify matrix Width x Height.
Colors look incorrect (e.g. Red is Green)	Color channel order mismatch	Change Color Order from RGB to GRB or BGR .
Alternating rows look backward or wavy	Serpentine (zigzag) vs Progressive layout mismatch	Switch Matrix Wiring Layout from Progressive to Serpentine .
Image is upside down or mirrored	Start corner or rotation angle mismatch	Use the Start Corner selector or click  Rotate 90° .